simplest turbulent flows starting from the first principles of mechanics like Newton's laws, without ever appealing to experimental data about the flow itself. For example, it's not currently possible to predict the pressure loss in a pipe in which the flow is turbulent, but it's known thanks to clever use of data at some level obtained in experiments. The basic issue is that the turbulent flow problems we are interested in are almost always highly non-linear, and the mathematics for handling such highly non-linear problems does not seem to exist. There has been a widespread view among many physicists that whenever a new problem arises in their subject, it appears that somehow, like magic, the mathematics required for it is found to have already been invented. The problem of turbulence provides a counter-example to this rule. The laws governing the problem are well known and, for simple incompressible fluids under normal conditions, are embodied in

DID YOU KNOW?

Nobel laureate Werner Heisenberg made brilliant advancements in Quantum Mechanics and is most well known for "The Uncertainty Principle", but his doctoral thesis was on the topic of turbulence.

the Navier-Stokes equations. But the solutions remain unknown. It is clear therefore that current mathematics is singularly ineffective in solving the problem of turbulence. As Richard Feynman put it, turbulence remains the greatest unsolved problem in classical Physics."

Such is the importance of turbulence studies that a whole new generation of computing techniques have spawned in its wake. Solving, or even coming close to a theory of turbulence will allow science to make better weather predictions, design power efficient automobiles and aircraft and have a better understanding of various natural phenomena.

The Origin of Life

We've always been obsessed with studying about the possibility of life on other planets, but another question that has baffled scientists even more is how exactly

A graphic rendition of a black hole. A theory of quantum gravity can help understand black holes better

did life as we know it originate on Earth? Though answering this question will have little practical application, the path to the answer could lead to several interesting discoveries in fields varying from microbiology to astrophysics.

Scientists believe that the key to understanding the origin of life could be in finding out how two of life's characteristic features – reproduction and genetic transfer – could have begun as processes involving molecules that have the ability to replicate. This led to the formation of the popular 'primordial soup' theory, according to which the early Earth somehow had a mixture, a sort of broth of molecules, that were energised by solar energy and electrical storms. Over a long period of time, these molecules would have reacted to form the more complex organic structures that make up life. This theory was buoyed by the famous Miller-Urey experiment, wherein the duo succeeded in creating amino acid by passing electrical discharges through a blend of simpler elements such as methane, ammonia, water and hydrogen. The discovery of DNA and RNA, however, doused the initial excitement as it seemed impossible that the complex and elegant structure of DNA could have risen from a primitive soup of chemicals.

A school of thought exists that believes that the early world was an RNA world rather than a DNA one. RNA was found to have the ability to speed up reactions and remain unchanged and to replicate and store genetic material. But in order for RNA to be hailed as the original replicator of life instead of DNA, scientists would have to find evidence of elements that could form nucleotides – the building blocks of the RNA molecule. The glitch is that nucleotides are extremely difficult to produce, even in a lab's controlled environment. Again, a primordial soup seems incapable of creating these molecules. This conundrum led to another school of thought that believed that the essential organic molecules present in primitive life have an extra-terrestrial origin and were shipped to Earth from outer space via comets, leading to the development of the 'panspermia' theory. Another possible explanation could be the 'Iron-Sulphur world' theory that proposes that life on earth has a deep sea origin, spawning from chemical reactions that occur in highly pressurised hot water found near hydrothermal vents with a volcanic nature.

It's quite remarkable that even after a good 200-odd years post industrialisation we're still not close to finding out how our Earth gave birth to life. There's been massive interest in this problem from research groups in interdisciplinary

DID YOU KNOW?

Discoverer of DNA's structure, Francis Crick, was the proponent of a theory known as 'directed panspermia', which hypothesised that advanced extraterrestrial civilisations deliberately seeded Earth with life.

sciences, resulting in several initiatives to carry out directed research in solving one of science's biggest mysteries.

Protein Folding

A trip down memory lane will take us back to high school chemistry/physics class where we all learnt (mostly by rote) that proteins are extremely important molecules and are the building blocks of life. Protein molecules are made up of amino acid sequences that influence their structure and, in turn, determine